

Production Planning and Execution (PP)

This case study explains an integrated production planning and execution process in detail and thus fosters a thorough understanding of each process step and underlying SAP functionality.

Product

S/4HANA 2020
Global Bike

Fiori 3.0

Level

Undergraduate
Graduate
Beginner

Focus

Production Planning and
Execution

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Version

4.1

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MOTIVATION

The data entry requirements in the production planning exercises (PP 1 through PP 6) were minimized because much of the data already existed in the SAP system. This stored data, known as master data, simplifies the processing of business transactions. Examples for this, are material master data, bills of materials, and routings.

In this case study, we will create consumption values for a finished product to plan and process a complete manufacturing cycle.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to successfully work through this case study, it is not necessary to have finished the PP exercises (PP 1 through PP 6). However, it is recommended.

NOTES

This case study uses the Global Bike data set.



 Process Overview

Learning Objective Understand and perform a manufacturing process cycle.

Time 200 min

Scenario In order to experience a complete manufacturing process you will take on different roles within the Global Bike Group, e.g., production supervisor, shop floor worker and plant manager. Overall, you will be working in the Materials Management (MM) and the Production Planning and Execution (PP) departments.

Employees involved

- Jun Lee (Production Manager)
- Hiro Abe (Plant Manager)
- Lars Iseler (Shop Floor Worker 2)
- Susanne Castro (Goods Receipt Clerk)
- Sanjay Datar (Warehous Employee)
- Michael Brauer (Shop Floor Worker 4)
- Jamie Shamblin (Controller)

Before you can start forecasting demand for your touring bike product group, changes in the material master record of the bikes need to be maintained.

Afterwards you will create a 12-month sales and operations plan (SOP) for your product group, receive the production relevant goods from the warehouse storage location and issue them to the production order.

To conclude the process, the production is confirmed as complete, the finished goods are received into the warehouse and costs assigned to the production order are analyzed.

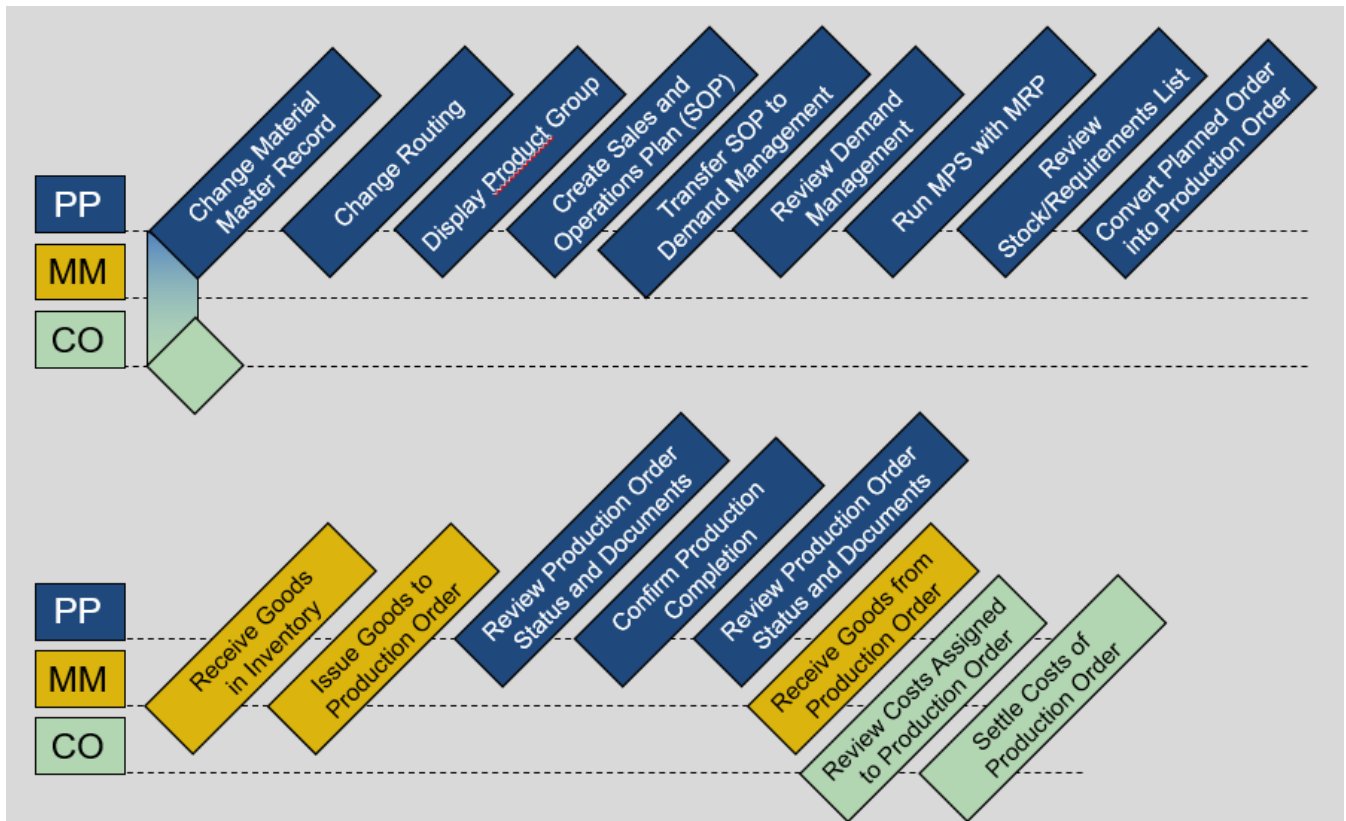


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Step 1: Change Material Master Record

Task Prepare a material master record for Demand Planning.

Time 20 min

Short Description In order to plan Global Bike’s deluxe touring bikes (black, silver and red) prepare their material master records by adding planning-relevant data to these records.

Name (Position) Jun Lee (Production Manager)

To change the views of a material, use the *Manage Product Master Data* app in the *Production Planning and Execution* area.

Manage Product Master Data

In the search screen, enter **DXTR*###** (replace ### with your three-digit number) in the search field.

DXTR*###

<

SAP

Manage Product Master Data

>

Standard*

▼

DXTR*000

✕

🔍

Editing Status:

All

▼

Product:

🔍

Product Description:

🔍

GTIN:

🔍

Product Group:

🔍

Product Category:

🔍

Adapt Filters (1)

Go

Press **Go**. Your various Deluxe Touring Bikes will be displayed.

Products (3)						Create	Copy	Mass Processing	Show in Hierarchy	Hide Draft Values	Delete	⚙️	📄	▼
☐	Image	Description / ID	Group / Type	GTIN	Product Category	Last Changed								
☐		Deluxe Touring Bike (black) DXTR1100	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:36:14 Chris Reich								
☐		Deluxe Touring Bike (silver) DXTR2100	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:36:58 Chris Reich								
☐		Deluxe Touring Bike (red) DXTR3100	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:37:40 Chris Reich								

Click the line of **Deluxe Touring Bike (red)** (DXTR3##) to open the details of the product.

<

SAP

Product

>

Deluxe Touring Bike (red)
DXTR3100

Edit

Copy

Show in Hierarchy

Product Type: Finished Product (FERT)

Product Category: Product

Product Group: Finished Bikes (BIKES)

Base Unit of Measure: Each (EA)

GTIN:

GTIN Category:

Revision Level:

General Information

Product Compliance

Components

Texts

Sales

Storage

Warehouse Management

Basic Data

Division:
Bicycles (BI)

Old Product Number:
—

Batch Management Required:
No

Marked for Deletion:
No

Created By:
Chris Reich

Created On:
08/19/2021, 11:51:26

Last Changed By:
Chris Reich

Last Changed On:
08/23/2021, 12:37:40

Press

Edit

 to switch to the edit mode.

Use the pull-down menu

>

 to select the *Plants* section. The window automatically scrolls to the correct position.

Plants

Attachments - Generic Object Services

General Information

Basic Data

Descriptions

Base Unit of Measure

...

Extended Service Parts Planning

Distribution Chains

Plants

Valuation Areas

Attachments - Document Management Services

You will see a list of all plants for which the product has been defined.

Plants

Search

Copy

Create

Delete

Plant	MRP Type	MRP Controller	Availability Check	Marked for Deletion
☐ DL00	M1	000	02	☐
☐ HD00	M1	000	02	☐
☐ HH00	M1	000	02	☐
☐ MI00	M1	000	02	☐
☐ SD00	M1	000	02	☐

>

1.

type -

40

Periods

<

SAP

Plant ▾

Deluxe Touring Bike (red) /

Plant Dallas

DL00

Advanced Planning ▾Extended Service Parts Planning ▾Forecasting ▾

Forecasting

Forecasting

Required Periods

Control Data

In the *Periods for Initialization* field enter **12**. Scroll down to the next area *Control data*.

12

In the *Control Data* area below, click on the value help icon  of the *Optimization Level* field and select the optimization level **F - Fine** (high optimization level). Then select **Parameter Optimization**.

F

Parameter Optimization

Now assign the smoothing factors. Enter **0.20** for the *Alpha Factor* (base value), **0.10** for the *Beta Factor* (trend value), **0.30** for the *Gamma Factor* (seasonal index) and **0.30** for the *Delta Factor* (MAD).

0,20
0,10
0,30
0,30

Required Periods

Historical Periods:

120

MON

Forecast Periods:

12

MON

Periods per Seasonal Cycle:

12

MON

Periods for Initialization:

12

MON

Fixed Periods:

0

MON

Control Data

Initialization Indicator:

Initialization by system (X)



Model Selection Indicator:



Optimization Level:

Fine (high optimization level) (F)



Tracking Limit:

4.000

Model Selection Procedure:

Analytical model selection procedure (2)



Weighting Group:



Reset Forecast Model Automatically:

☐

Parameter Optimization:

☒

Correction Factors:

☐

Alpha Factor:

0.20

Beta Factor:

0.10

Gamma Factor:

0.30

Delta Factor:

0.30

Select  to save the plant-specific data for plant DL00.

Click  to save your changes to the red Deluxe Touring Bike.

The SAP system updates the master data record for material DXTR3### and displays a corresponding message.

Master data record saved.

Select  to return to the Manage Product Master Data screen.

Repeat the same procedure for the silver and black Deluxe Touring Bike. Start with the silver one (**DXTR2###**) and then change the black bike (**DXTR1###**).

DXTR2###
DXTR1###

Click  to return to the SAP Fiori Launchpad.



Step 2: Change Routing

Task Change a routing for a finished good.

Time 15 min

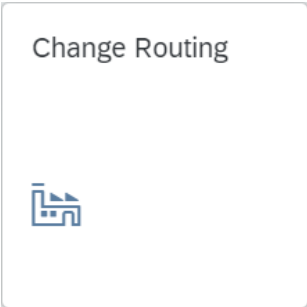
Short Description Change the routing for your red Deluxe Touring bike.

Name (Position) Jun Lee (Production Manager)

After the operational steps are laid out, the components must be allocated to the individual operations. This is a progressive process where each operation builds off the materials that entered production in the previous operations.

To change the routing, use the *Change Routing* app in the *Production Planning and Execution* area.

Change Routing



Enter the material number of your red Deluxe Touring bike (**DXTR3###**). In the *Plant* field, enter the Global Bike plant number in Dallas (**DL00**).

DXTR3###
DL00

< SAP Change Routing: Initial Screen

Routings Sequences Operations More

Material: DXTR3100

Plant: DL00

Sales document: Sales Document Item:

WBS Element:

Group:

Also make sure that the *Planner Group* field is empty. Then press **Operations**.

< **SAP**

Change Routing: Operation Overview

< > More >

Group: 1804

Group Counter: 1

Material: DXTR3100

Deluxe Touring Bike (red)

Operation Overview

	Op...	SOp	Work cen...	Plant	* C...	Standard...	Description	Lo...
☐	0010		ASSY1000	DL00	ASSY		Material staging	☐
☐	0020		ASSY1000	DL00	ASSY		Attach seat to frame	☐
☐	0030		ASSY1000	DL00	ASSY		Attach handle bar assembly	☐
☐	0040		ASSY1000	DL00	ASSY		Attach derailleur gear asm. to wheel	☐
☐	0050		ASSY1000	DL00	ASSY		Attach front and rear wheels to chain	☐
☐	0060		ASSY1000	DL00	ASSY		Attach brakes	☐
☐	0070		ASSY1000	DL00	ASSY		Attach peddles	☐
☐	0080		INSP1000	DL00	ASSY		Test bike	☐
☐	0090		PACK1000	DL00	ASSY		Disassemble	☐
☐	0100		PACK1000	DL00	ASSY		Pack bike	☐
☐	0110		PACK1000	DL00	ASSY		Move to storage	☐

Note A routing is defined by the routing group and the routing group counter. Moreover, the routing contains a reference to the material whose production is described by the routing.

Besides the standard sequence, it can also have parallel or alternative sequences. Alongside the standard values, the routing also contains time elements that are relevant for scheduling operations. Each operation in the routing may contain its own base quantity, to which these time elements may refer.

Select **Allocation** to display a list of all components. If this is not displayed, you will find the entry in the pull-down menu under **More ▶ Allocation**.

Select the rows Touring Frame-Red (**TRFR3##**) and Touring Seat Kit (**TRSK1##**).

Item Overview

	Ph...	Le...	Path	Ite...	Component	Quantity
☐	☐	0	0	0010	TRWA1100	2
☒	☐	0	0	0020	TRFR3100	1
☐	☐	0	0	0030	DGAM1100	1
☒	☐	0	0	0040	TRSK1100	1
☐	☐	0	0	0050	TRHB1100	1
☐	☐	0	0	0060	PEDL1100	1
☐	☐	0	0	0070	CHAN1100	1
☐	☐	0	0	0080	BRKT1100	1
☐	☐	0	0	0090	WDOC1100	1
☐	☐	0	0	0100	PCKG1100	1

Press **New Assignment**. In the popup that appears, enter **0020** for Activity and confirm the entry with .

TRFR3###
TRSK1###

0020

New Assignment

Assign to

Activity: 0020Sequence: 0

✓

Oper./act. list

🔍

✗

Back in the *Material Component Overview* you can see that both components have now been assigned to *Activity 0020*.

Item Overview											
	Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity Seq.
☐	☐	0	0	0010	TRWA1100	2		EA	L		
☐	☐	0	0	0020	TRFR3100	1		EA	L	☐	0020 0
☐	☐	0	0	0030	DGAM1100	1		EA	L		
☐	☐	0	0	0040	TRSK1100	1		EA	L	☐	0020 0
☐	☐	0	0	0050	TRHB1100	1		EA	L		
☐	☐	0	0	0060	PEDL1100	1		EA	L		
☐	☐	0	0	0070	CHAN1100	1		EA	L		
☐	☐	0	0	0080	BRKT1100	1		EA	L		
☐	☐	0	0	0090	WDOC1100	1		EA	L		
☐	☐	0	0	0100	PCKG1100	1		EA	L		

Repeat this process for all other components and assign them to the operations below.

Component	Operation
TRHB1### (touring handlebar)	0030
TRWA1### (touring aluminum wheel assembly)	0040
DGAM1### (derailleur gear assembly)	0040
CHAN1### (chain)	0050
BRKT1### (brake kit)	0060
PEDL1### (pedal assembly)	0070
WDOC1### (warranty document)	0100
PCKG1### (packaging)	0100

- TRHB1### - 0030
- TRWA1### - 0040
- DGAM1### - 0040
- CHAN1### - 0050
- BRKT1### - 0060
- PEDL1### - 0070
- WDOC1### - 0100
- PCKG1### - 0100

Item Overview											
	Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity Seq.
☐	☐	0	0	0010	TRWA1100	2		EA	L	☐	0040 0
☐	☐	0	0	0020	TRFR3100	1		EA	L	☐	0020 0
☐	☐	0	0	0030	DGAM1100	1		EA	L	☐	0040 0
☐	☐	0	0	0040	TRSK1100	1		EA	L	☐	0020 0
☐	☐	0	0	0050	TRHB1100	1		EA	L	☐	0030 0
☐	☐	0	0	0060	PEDL1100	1		EA	L	☐	0070 0
☐	☐	0	0	0070	CHAN1100	1		EA	L	☐	0050 0
☐	☐	0	0	0080	BRKT1100	1		EA	L	☐	0060 0
☐	☐	0	0	0090	WDOC1100	1		EA	L	☐	0100 0
☐	☐	0	0	0100	PCKG1100	1		EA	L	☐	0100 0

Apply your changes with **Save**. The system issues a message that the routing has been saved.

✓ Routing was saved with group 1804 and material DXTR3100.

Click  to return to the SAP Fiori Launchpad.



Step 3: Display Product Group

Task Display a product group.

Time 5 min

Short Description Display the product group (product family) for all your Deluxe Touring bikes.

Name (Position) Jun Lee (Production Manager)

A product group (product family) supports high-level planning. This way, it is not necessary to delve into the minutia of creating planning forecasts for every material in the company.

Product group

To view the Deluxe Touring bike product group, use the *Display Product Group* app in the *Production Planning and Execution* area.

Display Product Group



In the *Display Product Group: Initial Screen*, in the *Product group* field find and select your group for deluxe touring bikes. In order to do so, press the search icon  (or pressed F4), enter **###*** in the *Material description* field. Remember to replace **###** with your three-digit number, e.g., enter 009* if your number is 009. Enter **DL00** as *Plant*.

###*

DL00

Name of the product group (1) ✕

Find product group via MRP controller

Find product group via description

Find product group via plant

MRP Controller:

Material description: 000*

Language Key: EN

Product group:

Plant: DL00

Maximum No. of Hits: 500

Find

Multiple Selection

Close

Then, press Enter or click on  to display the search results.

Name of the product group (1)

> Find product group via MRP controller

Find product group via description

Find product group via plant

MRP Controller	Material description	Language
☐	000 PRODUCT GROUP BICYCLES	EN
☒	000 PRODUCT GROUP DELUXE TOURING BICYCLE	EN
☐	000 PRODUCT GROUP OFFROAD BICYCLES	EN
☐	000 PRODUCT GROUP PROFESSIONAL TOURING	EN
☐	000 PRODUCT GROUP TOURING	EN

5 Entries found

You will see a list of all your product groups, e.g., for mountain bikes or touring bikes. Select the group of Deluxe Touring Bikes (**PG-DXTR##**). Then click  to apply the selection.

PG-DXTR###

Now that the correct *Product group* (**PG-DXTR###**) is filled in, make sure that *Plant* **DL00** is entered.

DL00

< SAP Display Product Group: Initial Screen

More ▾

Product group: PG-DXTR000

Plant: DL00

Then, press Enter to display the product group details.

On this screen, you can see that this product group defines proportions for three different bikes: the black, silver and red deluxe touring bike. For the black bike, a share of 40% will be considered and 30% for the silver and the red bikes each.

<  Display Product Group: Members (Materials)

[Next Level](#) [Hierarchy Graphic](#) [Versions...](#) [Master Data...](#) [Product Group Graphic](#) [More](#) 

Product group: PG-DXTR000

000 Product Group Deluxe Touring Bicycle

Plant: DL00 : Plant Dallas

Base Unit: EA

Member number	Plnt	Unit conv.	Aggr.fact.	Proportion	UoM	V	M	Fx	Short Text
DXTR1000	DL00	1	1	40	EA			☐	Deluxe Touring Bike (black)
DXTR2000	DL00	1	1	30	EA			☐	Deluxe Touring Bike (silver)
DXTR3000	DL00	1	1	30	EA			☐	Deluxe Touring Bike (red)
								☐	

Click  to return to the SAP Fiori Launchpad.



Step 4: Create Sales and Operations Plan (SOP)

Task Create a sales and operations plan for a product group.

Time 20 min

Short Description Create a 12-month sales and operations plan (SOP) for your product group.

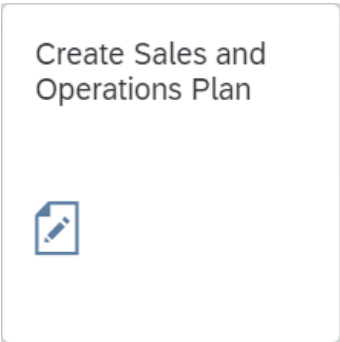
Name (Position) Jun Lee (Production Manager)

A sales and operations plan (SOP) is a planning tool that is used to consolidate data for forecasting future sales and production levels as well as the methods needed to meet those requirements. In this task, our SOP will be based on historical consumption values taken from a fixed period. This is in contrast to forecasting within a real-life system which would base the prediction on previous periods and their respective consumption.

Sales and operations plan

To create an SOP, use the *Create Sales and Operations Plan* app in the *Production Planning and Execution* area.

Create Sales and Operations Plan



Make sure that *Product group* **PG-DXTR###** and *Plant* **DL00** are entered. Then, select **Active version**.

PG-DXTR###
DL00

Record the version number: _____

In the menu bar, select:

More ▶ Edit ▶ Create sales plan ▶ Forecast...

Select **Period intervals**, *Forecast* from **current period/current year** to **previous period/next year**, *Historic Data* from **06/2017** to **03/2021**, *Forecast execution* **Aut. model selection**. Compare your screen with the one below before clicking on **Historical...** to view the historical values.

Period intervals
current period/current
year
previous period/next
year
04/2017
03/2021
Aut. model selection

Forecast: Model Selection ✕

Periods

☒ Period intervals

Forecast * Fr... : 06/2022 * To : 05/2023

Historical data * Fr... : 04/2017 * To : 03/2021

☐ No. of periods

No. of forecast periods:

No. of historical values:

Forecast execution

☐ Constant models ☐ Seasonal models

☐ Trend models ☐ Season. Trend Models

☒ Aut. model selection ☐ Historical

Forecast parameters

Profile:

[Forecasting](#)
[Historical...](#)
[Forecast profile...](#)
[Version...](#)
✕

You will get an overview of the passed periods in the specified time range.

Forecast: Historical Values ✕

Historical values

Period	Val. fld	Corr.value	F	C
M 03/2021	333	333	☐	☐
M 02/2021	340	340	☐	☐
M 01/2021	363	363	☐	☐
M 12/2020	310	310	☐	☐
M 11/2020	276	276	☐	☐
M 10/2020	283	283	☐	☐
M 09/2020	306	306	☐	☐
M 08/2020	283	283	☐	☐

[Forecasting](#)
[Correct](#)
✕

Click on Forecasting.

In the next Popup the system selected *Trend and season*. Click on Forecasting again.

In the next pop-up you can see that the system tested and found Seasonal and Trend tendencies in the past consumption data and has applied a Seasonal Trend Model.

Forecast: Results ✕

Basic value:
 MAD:

Trend value:
 Error total:

Forecast results

Period	Orig. HV	Corr. HV	Ex-post FV	Orig. FV	Corr. FV	Season	F	C
M 04/2021				317	317	0.97	☐	☐
M 05/2021				342	342	1.03	☐	☐
M 06/2021				291	291	0.87	☐	☐
M 07/2021				307	307	0.90	☐	☐
M 08/2021				344	344	0.99	☐	☐
M 09/2021				383	383	1.09	☐	☐
M 10/2021				347	347	0.97	☐	☐

No forecast error messages exist

✓ 📄 🕒 Forecasting 🔗 User exit

Press , the sales forecast has been transferred to the SOP.

Look at the planning table. Note that your values may differ from the screenshot.

< **SAP** Change Rough-Cut Plan 🔍

Characteristic 📄 More ⌵

Product group: 000 Product Group Deluxe Touring Bicycle
 Plant:
 Version: Active version Active

SOP: plan individual product group

Planning Table	Un	M 06/2022	M 07/2022	M 08/2022	M 09/2022	M 10/2022	M 11/2022
☒ Sales	EA	346	364	407	452	408	395
☐ Production	EA						
☐ Stock level	EA	-346	-710	-1118	-1570	-1979	-2375
☐ Target stock level	EA						
☐ Range of Coverage							
☐ Target days' supply							

As *Target day's supply* enter 5 for each forecasted period.

5

SOP: plan individual product group

Planning Table	Un	M 08/2022	M 09/2022	M 10/2022	M 11/2022	M 12/2022	M 01/2023
☒ Sales	EA	407	452	408	395	459	505
☐ Production	EA						
☐ Stock level	EA	-1117	-1569	-1977	-2372	-2831	-3336
☐ Target stock level	EA						
☐ Range of Coverage							
☐ Target days' supply		5	5	5	5	5	5

In a production plan, you plan the quantities you need to produce in order to meet your sales plan. The system then calculates stock levels and days' supply for each period on the basis of the sales and production quantities and any target data. There are several different planning strategies available, which differ in the production values and the stock levels proposed.

As the SOP is a high-level planning, discrete production values are not necessary. The SAP system calculates discrete numbers once the SOP is transferred to the Demand Management.

In the menu bar, select:

More ► Edit ► Create production plan ► Synchronous to sales

Note the change in the *Production* and in the *Stock* level rows. The production plan is created to match the sales forecast.

SOP: plan individual product group

Planning Table	Un	M 08/2022	M 09/2022	M 10/2022	M 11/2022	M 12/2022	M 01/2023
☐ Sales	EA	407	452	408	395	459	505
☐ Production	EA	407	452	408	395	459	505
☐ Stock level	EA						
☐ Target stock level	EA						
☐ Range of Coverage							
☐ Target days' supply		5	5	5	5	5	5

In the system menu, select

More ► Edit ► Create production plan ► Target day's supply

Note the impact on the production plan and stock levels. Production levels are generated to match the sales plus the amount needed to be put into stock to meet the target days' supply specifications.

Review the Planning Table (your numbers may be different).

SOP: plan individual product group

Planning Table	Un	M 08/2022	M 09/2022	M 10/2022	M 11/2022	M 12/2022	M 01/2023
☐ Sales	EA	407	452	408	395	459	505
☐ Production	EA	413	461	398	395	467	512
☐ Stock level	EA	65	75	65	65	74	81
☐ Target stock level	EA						
☐ Range of Coverage		5	5	5	5	5	5
☐ Target days' supply		5	5	5	5	5	5

Note Although the screen displays integer production values, the SAP system calculates with decimal precision. You can display the decimal places of a series by pressing F8. Then create the production plan.

Accept the SOP with . A system message appears and you return to the initial screen.

 Plan saved under version number A00

Click  to return to the SAP Fiori Launchpad.



Step 5: Transfer SOP to Demand Management

Task Transfer SOP to Demand Management.

Time 10 min

Short Description Transfer the Sales and Operations Plan to Demand Management.

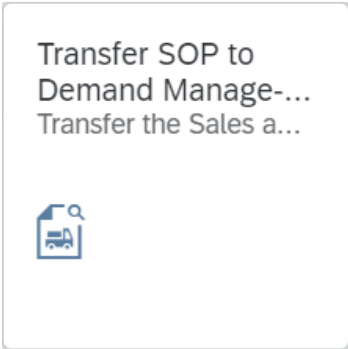
Name (Position) Jun Lee (Production Manager)

Demand Management is the tool used to disaggregate planning data from high-level plans down to the detailed planning level. For this task, planning for the Deluxe Touring Product Group will be broken down into the individual components that belong to this group.

Demand Management

To transfer sales/rough planning to Demand Management, use the *Transfer Planning Data to Demand Management* app in *Production Planning and Execution* area.

Transfer Planning Data to Demand Management



Enter *Product group* **PG-DXTR###**, *Plant* **DL00**, and the version saved in the previous task (**A00**).

PG-DXTR###
DL00
A00

Select **Prod.plan for mat. or PG members as proportion of PG** and **Active**. Then, deselect the **Invisible transfer** indicator to present the disaggregation results on another screen allowing the planner to modify the results before saving them manually to Demand Management.

Prod.plan for mat. or PG
members as prop. of PG
Active
~~Invisible transfer~~

<

SAP

Transfer Planning Data to Demand Management

Transfer now More ▾

* Product group:

PG-DXTR000

000 Product Group Deluxe Touring Bicycle

* Plant:

DL00

Plant Dallas

Version:

A00

Transfer strategy and period

☐ Sales plan for material or PG members

☐ Sales plan for mat. or PG members as proportion of PG

☐ Production plan for material or PG members

☒ Prod.plan for mat. or PG members as proportion of PG

From:

06/26/2022

To:

☐ Invisible transfer

Independent requirement specifications

Requirements type:

Version:

☒ Active

Select **Transfer now** and examine the Planned Independent Requirements generated for **DXTR1###**.

<

SAP

Plnd Ind. Reqmts: Planning Table

🔍

🔊 ⓘ ⚙️ Compare 🗑️ Delete ⋮ History 📄 📄 📄 📄 📄 More ▾

Planning start: 06/26/2022

Planning End:

Table

Items

Schedule Lines

<< < > >>

Material	MRP ...	V	A	BU	Reqmnt Segment	M 06/2022	M 07/2022	M 08/2022	M 09/2022	M 10/2022
☐ DXTR1000	☐ 00	AG	☒	EA		161	146	165	184	159

Then click on **Save** to save the demand for the DXTR1###.

After saving, the system jumps to the independent requirement of the next material (DXTR2###). Now examine the independent requirement generated for DXTR2###.

<< < > >>

Material	MRP ...	V	A	BU	Reqmnt Segment	M 06/2022	M 07/2022	M 08/2022	M 09/2022	M 10/2022
☐ DXTR2000	☐ DL00	AG	☒	EA		121	110	124	138	119

Continue with **Save**. Finally, examine the planned independent requirement of material DXTR3###.

<<<<>>>>										
Material	MRP ...	V	A	BU	Reqmnt Segment	M 06/2022	M 07/2022	M 08/2022	M 09/2022	M 10/2022
☐ DXTR3000	☐ 00	AG	☒	EA		121	110	124	138	119

Also save this requirement with

Save

.

Note DXTR1### makes up 40%, DXTR2### makes up 30% and DXTR3### another 30% of the production plan created in your Sales and Operations Plan.

You will automatically return to the initial screen. The system also provides you with a message that the requirement has been saved. You can also see this in the other materials in advance.



Click to return to the SAP Fiori Launchpad.



Step 6: Review Demand Management

Task Review the requirements for a product group.

Time 10 min

Short Description Review the requirements for the product group to ensure that there are production requirements for the individual production items.

Name (Position) Hiro Abe (Plant Manager)

To view planned requirements, use the *Maintain PIRs* app in *Production Planning and Execution* area.

Maintain PIRs

A welcome message is displayed informing you that you do not yet belong to any area of responsibility. Confirm this with . You will be forwarded automatically.

The *My Area of Responsibility* screen appears with a list of existing plants. Set your responsibility for the plant **DL00**.

DL00

< SAP

My Area of Responsibility ▾

Q ? TT

Standard* ▾
Not Filtered

Plant / MRP Controller Combinations (7) Assign Unassign ⚙️ 📄 ▾

☐	Plant	Plant Name	MRP Controller	MRP Controller Name	AOR Status
☐	0001		001	PERSON 1	
☐	0003		001	PERSON 1	
☐	DL00	Plant Dallas	000	DL MRP Controller	
☐	HD00	Plant Heidelberg	000	HD MRP Controller	

Then click  to return. In the *Maintain PIRs* screen expand the search by pressing .

SAP

Maintain PIRs

Standard*

Filtered By (3): Plant, Period Indicator, Version Active

Materials (9)

CreateUpload and EditEditShow Details

☐	Material	Plant	Reach	Accuracy - Current Period	Accuracy - Last Week	Version Active	Last Modified Date
☐	DGRB2000	Plant Dallas (DL00)	0 Months	100 %	0 %	Yes	09/03/2021
☐	DGRW2000	Plant Dallas (DL00)	1 Months	100 %	0 %	Yes	09/16/2021
☐	DXTR1000	Plant Dallas (DL00)	11 Months	99999 %	0 %	Yes	09/13/2021

The plant in Dallas (DL00) is already preselected. Now enter **DXTR*###** as the *search term* and change the *Version Active* field to **No** and select **Yes**.

DXTR*###
no
yes

SAP

Maintain PIRs

Standard*

Material:DXTR*110

Plant:Plant Dallas (DL00)

MRP Area:

Accuracy: Current Period:

Last Modified Date:MM/dd/yyyy - MM/dd/yyyy

Period Indicator:Monthly (M)

Reach:

Version Active:

Yes x

No

Yes

Adapt Filters (4)Go

CreateUpload and EditEditShow Details

Press **Go** to run the search with the new criteria. You will now be shown your three Deluxe Touring Bikes.

Materials (3)

CreateUpload and EditEditShow Details

☐	Material	Plant	Reach	Accuracy - Current Period	Accuracy - Last Week	Version Active	Last Modified Date
☐	DXTR1000	Plant Dallas (DL00)	11 Months	99999 %	0 %	Yes	06/26/2022
☐	DXTR2000	Plant Dallas (DL00)	11 Months	99999 %	0 %	Yes	06/26/2022
☐	DXTR3000	Plant Dallas (DL00)	11 Months	99999 %	0 %	Yes	06/26/2022

Select all three lines and click **Edit (3)**.

View planned independent demands for the Deluxe Touring Bike product group for all 3 materials.

SAP

Edit PIRs

Maintain Quantities per Period (3)

Jun 26, 2022

Material (Plant / MRP Area / Version / ReqType)	Version Is Ac...	ReqPlan	ReqSeg	UoM	M06.2022	M07.2022	M08.2022	M09.2022	M10.2022	M11.2022	M12.2022	M01.2023	M02.2023	M03.2023	M04.2023	M05.2023
DXTR1000 (DL00 / DL00 / AG / VSP)	YES			EA	181	146	195	184	159	158	187	205	184	174	177	190
DXTR2000 (DL00 / DL00 / AG / VSP)	YES			EA	121	110	124	138	119	119	140	154	138	131	133	143
DXTR3000 (DL00 / DL00 / AG / VSP)	YES			EA	121	110	124	138	119	119	140	154	138	131	133	143

Click **SAP** to return to the SAP Fiori Launchpad.

Step 7: Run MPS with MRP

Task Run Master Production Scheduling (MPS).

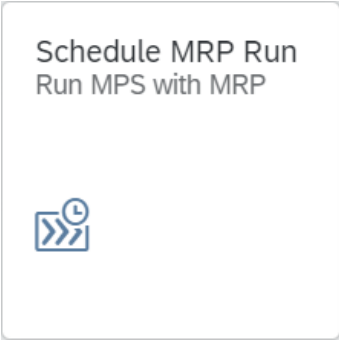
Time 10 min

Short Description Run Master Production Scheduling (MPS) to generate a series of planned orders that satisfy the requirements from SOP and demand management. Concurrently with MPS, the MRP materials will be processed leading to the generation of planned orders for dependent requirements that have been created by the BOM explosion process.

Name (Position) Jun Lee (Production Manager)

To start Master Production Scheduling, use the *Schedule MRP Run - Run MPS with MRP* app in the *Production Planning and Execution* area.

Schedule MRP Run



Enter your *Material* **DXTR3###**, and as *Plant* **DL00**.

DXTR3###
DL00

The control parameters can be adopted and should be filled in by the system as follows:

- *Processing Key:* **NETCH** (Net-Change in Total Horizon)
- *Create Purchase Req.:* **2** (Purchase requisition in opening period)
- *SA Deliv. Sched. Lines:* **3** (Schedule lines)
- *Create MRP List:* **1** (MRP list)
- *Planning mode:* **1** (Adapt planning data (normal mode))
- *Scheduling:* **1** (Determination of Basic Date for Planned)

NETCH
2
3
1
1
1

Then, select **Display material list**.

Display material list

< **SAP** Single-Item, Multi-Level

More ▾

* Material: DXTR3000

MRP Area:

Plant: DL00

Scope of Planning

☐ Product group

MRP Control Parameters

* Processing Key: NETCH	Net Change in Total Horizon
* Create Purchase Req.: 2	Purchase requisitions in opening period
* SA Deliv. Sched. Lines: 3	Schedule lines
* Create MRP List: 1	MRP list
* Planning mode: 1	Adapt planning data (normal mode)
* Scheduling: 1	Determination of Basic Dates for Planned

Process Control Parameters

☐ Also Plan Unchanged Components

☐ Display Results Prior to Saving

☒ Display material list

☐ Simulation mode

Press Enter. A warning message will appear asking you to check input parameters. Press Enter to confirm and bypass the warning message.

Note In MRP, a net requirements calculation is executed in the planning run to determine whether a material shortage exists for a certain material. In addition, stock and fixed receipts that currently exist (for example, purchase orders, production orders, fixed purchase requisitions and planned orders) are compared with the safety stock and requirements. The result of this comparison is the quantity available for planning.

If the quantity available for planning is lower than zero, a material shortage exists. MRP reacts to material shortages by creating new procurement proposals (purchase requisitions and planned orders). The suggested procurement quantity results from the lot-sizing procedure that is set in the material master.

As soon as the planning run is completed, a result overview is displayed. Check the planning details of the result overview.

SAP Single-Item, Multi-Level	
Materials	More
Statistics	
Materials planned	17
Materials with New Exceptions	17
Materials with Termination MRP List	
Parameters	
MRP Area	DL00
Plnt	DL00
Processing Key	NETCH
Create Purchase Requisition	2
SA Schedule Line	3
Create MRP List	1
Planning Mode	1
Scheduling	1

Scroll down further, there you will see a detailed listing of all materials considered.

Ranking List of Materials with Highest CPU Times (in ms)					
Material	Runtime		MRP Area		Plnt
	Read	Net Calc.	BOM	LdTmeSched	Update
DXTR3000	9,868	993	579	DL00	DL00
			411	0	1,206
BRKT1000	1,353	8	1,233	DL00	DL00
			0	0	36
BOLT1000	421	2	15	DL00	DL00
			0	0	402
TRWA1000	285	2	11	DL00	DL00
			140	0	131
TRWH1000	77	2	10	DL00	DL00
			0	0	64
TRTR1000	60	2	10	DL00	DL00
			0	0	47
LWSH1000	59	2	9	DL00	DL00
			0	0	47
PEDL1000	56	2	21	DL00	DL00
			0	0	32
HXNT1000	54	2	9	DL00	DL00
			0	0	42
CHAN1000	51	2	16	DL00	DL00
			0	0	32

Click  to return to the SAP Fiori Launchpad.



Step 8: Review Stock/Requirements List

Task Review the Stock/Requirements List.

Time 10 min

Short Description Review the Stock/Requirements List for your deluxe touring bike.

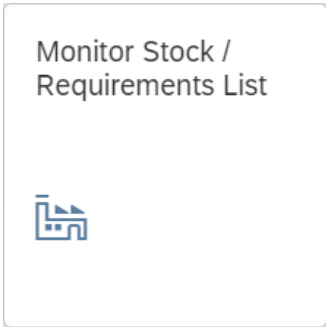
Name (Position) Lars Iseler (Shop Floor Worker 2)

The Stock/Requirements List is a list which dynamically changes whenever a transaction occurs using the given material. Display and review the Stock/Requirements List for all materials of the red deluxe touring bike on hand and the demand that exists against these products. The report shows that there is no stock and therefore nothing is available for use at this time.

Stock/Requirements List

To display the stock/requirements list, use the *Monitor Stock/Requirements List* app in the *Production Planning and Execution* area.

Monitor
Stock/Requirements List



On the *Individual access* tab, enter *Material DXTR3###* and *Plant DL00*. Click on **Continue** to display the associated stock/requirements list.

DXTR3###
DL00

< **SAP** Stock/Requirements List: Initial Screen

More ▾

Individual access

Collective access

* Material: DXTR3000

Description: Deluxe Touring Bike (red)

MRP Area:

Plant: DL00 Plant Dallas

With filter: ☐

Currently, the system lists all entries as single rows.

Material:

Description:

MRP Area:

Plant Dallas

Ex. manuf.:

Plant:

MRP type:

Material type:

Unit:

Icon	A...	Date	MRP e...	MRP element data	Rescheduling...	E...	Receipt/Reqmt	Available Qty	Pro...
Icon		06/26/2022	Stock						0
Icon		06/01/2022	IndReq	VSF			121-	121-	
Icon		07/01/2022	IndReq	VSF			110-	231-	
Icon		07/03/2022	----->	End of Planning Tim...					
Icon		07/03/2022	PldOrd	0000000001/STCK	06/01/2022	30	121	110-	0001
Icon		07/03/2022	PldOrd	0000000002/STCK			110	0	0001
Icon		08/01/2022	PldOrd	0000000003/STCK			124	124	0001
Icon		08/01/2022	IndReq	VSF			124-	0	
Icon		09/01/2022	PldOrd	0000000004/STCK			138	138	0001
Icon		09/01/2022	IndReq	VSF			138-	0	
Icon		10/01/2022	PldOrd	0000000005/STCK			119	119	0001
Icon		10/01/2022	IndReq	VSF			119-	0	
Icon		11/01/2022	PldOrd	0000000006/STCK			119	119	0001
Icon		11/01/2022	IndReq	VSF			119-	0	
Icon		12/01/2022	PldOrd	0000000007/STCK			140	140	0001
Icon		12/01/2022	IndReq	VSF			140-	0	
Icon		01/01/2023	PldOrd	0000000008/STCK			154	154	0001
Icon		01/01/2023	IndReq	VSF			154-	0	
Icon		02/01/2023	PldOrd	0000000009/STCK			138	138	0001
Icon		02/01/2023	IndReq	VSF			138-	0	
Icon		03/01/2023	PldOrd	0000000010/STCK			131	131	0001
Icon		03/01/2023	IndReq	VSF			131-	0	

 Σ

Days Weeks Months							
Date GR ST On		Page		1 / 1			
A...	Period/Segment	Plnd ind.req...	Requirement	Receipts	Avail. Quantity	ATP quantity	Actual
	Stock				0	0	26.0
	06/01/22	121-	0	0	121-	0	33.0
	07/01/22	110-	0	0	231-	0	3.0
	07/03/22 End of Planni						
	07/03/22	0	0	231	0	231	0.0
	08/01/22	124-	0	124	0	124	0.0
	09/01/22	138-	0	138	0	138	0.0
	10/01/22	119-	0	119	0	119	0.0
	11/01/22	119-	0	119	0	119	0.0
	12/01/22	140-	0	140	0	140	0.0
	01/01/23	154-	0	154	0	154	0.0
	02/01/23	138-	0	138	0	138	0.0
	03/01/23	131-	0	131	0	131	0.0
	04/01/23	133-	0	133	0	133	0.0
	05/01/23	143-	0	143	0	143	0.0



✓ ⌘ ✎ ⚙ ⚙ ⓘ -> Prod.Ord. -> PartConvProdOrder -> Proc.Ord. -> SubProcOrd -> Pur Req. ...



Click  to return to the SAP Fiori Launchpad.



Step 9: Convert Planned Order into Production Order

Task Convert a planned order into a production order.

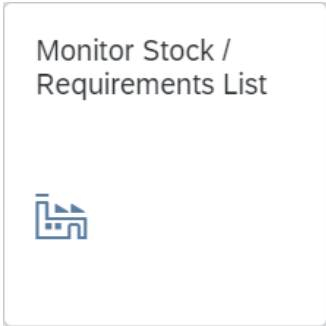
Time 10 min

Short Description Convert a planned order generated in the MPS/MRP run to a production order. The stock requirements list displays the suggested planned orders from the MPS run.

Name (Position) Lars Iseler (Shop Floor Worker 2)

To convert planned orders into production orders, use the SAP Fiori app *Monitor Stock / Requirements List* in the *Production Planning and Execution* area.

Monitor Stock / Requirements List



Enter **Material DXTR3###**, **Plant DL00**, and click on **Continue**.

DXTR3###
DL00

Material:

Description:

MRP Area: Plant:

Plant Dallas

Ex. manuf.:

MRP type: Material type: Unit:

Σ ↺ ↻ ↷ ↸ ↹ ↺ ↻ ↷ ↸ ↹

GR ST On On Vendor Cust.

A...	Date	MRP e...	MRP element data	Rescheduling...	E...	Receipt/Reqmt	Available Qty	Pro...
🔍	06/26/2022	Stock					0	
🔍	06/01/2022	IndReq	VSF			121 -	121 -	
🔍	07/01/2022	IndReq	VSF			110 -	231 -	
🔍	07/03/2022	----	End of Planning Tim...					
🔍	07/03/2022	PldOrd	0000000001/STCK	06/01/2022	30	121	110 -	0001
🔍	07/03/2022	PldOrd	0000000002/STCK			110	0	0001
🔍	08/01/2022	PldOrd	0000000003/STCK			124	124	0001
🔍	08/01/2022	IndReq	VSF			124 -	0	

Choose at the beginning of the **third** planned order. A popup opens with details about the order.

Additional Data for MRP Element ✕

Pld Order: Make-to-stock

Order End Date:

GR pr.time:

Order Qty: EA

Order Start:

Proc. Type:

Scrap:

Opening Date:

Order Type:

✓ ↺ ↻ ↷ ↸ ↹

[-> Prod.Ord.](#) [-> PartConvProdOrder](#) [-> Proc.Ord.](#) [-> SubProcOrd](#) [-> Pur Req.](#) ★ ✕

The planned order should now be converted into a production order. To do this, press **-> Prod.Ord.** . The system creates a temporary production order, identified by the generic order number, and releases it automatically.

Order: %00000000001

Material: DXTR3000

Status: REL MACM SETC

Deluxe Touring Bike (red)

General

Assignment

Goods Receipt

Control

Dates/Quantities

Master Data

Long Text

Administration

Items

Fast Er

Quantities

Total Qty: 124

EA

Scrap Portion: 0.00 %

Delivered: 0

Short/Exc. Rcpt: 0

Note At this point, please note down the total quantity in your production order. You will need it later when confirming your order.

Total quantity

Determine the status of your order by clicking on .

Status

Business processes

Syst. Status

X	Stat...	Text
☒	REL	Released
☒	MACM	Material committed
☒	SETC	Settlement rule created

Note When you converted the planned order to a production order, scheduling took place, an availability check was automatically carried out and a reservation was placed on the materials specified within the bill of materials.

Click on , to go back to the *Production order Create: Header* screen and save your production order with .

Note When you save the production order the system will automatically calculate the planned costs for the production order.

The system assigns a unique number to the production order. Please make a note of the production order number.

 Order number 1000000 saved

Production order
number

You will automatically return to the requirements/inventory list. Select  to refresh the list. The planned order **PldOrd** that you had selected is now available as a production order **PrdOrd**.

Step 10: Receive Goods in Inventory

Task Receive goods in the Dallas plant.

Time 10 min

Short Description Receive enough goods in the Dallas storage locations to start the production process.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Usually, at this point the purchasing department in Dallas would take over and procure enough raw materials from vendors to fill the inventory so that the production process can be initiated. In this case study, we are bypassing this procurement process (this process is explained in the MM unit in detail). Because the inventory for all DXTR3### components is empty, we will assume that we find 500 pieces each in the storage location.

Goods receipt

To receive goods in the inventory, use the app *Post Goods Receipt without Reference* in the *Production Planning and Execution* area.

Post Goods Receipt without Reference



The document and posting date are already defaulted with the current date and can thus be transferred.

Goods Receipt without Reference

General Information

Items

Attachments

Printing: No print

Note:

Document Date: 06/26/2022

Delivery Note:

Posting Date: 06/26/2022

Directly below you will find the *Item* section. The table there is ready for input and offers **item 01** in advance.

Items

Items

Copy

Delete

Create

☐ Item	Material	Quantity / Unit	Plant	Storage Location
☐ 01		0.000		

Stock Type:

Click on the line of **item 01**, you will switch to a separate input window.

Enter your *Material* **TRWA1###** and press Enter.

Now you can enter the *Quantity* **500** with *Unit* **EA**.

Next, choose *Plant* **DL00**. When choosing the *Storage Location* your storage for semi-finished goods is will be proposed. You can check the status directly in the selection screen.

TRWA1###
500
EA

DL00
SF00

Select Storage Location

Search

Standard

Semi-Fin. Goods

SF00

Unrestricted...

Quality

Blocked

0.000 EA

0.000 EA

0.000 EA

Choose the *Storage Location* **SF00**. The *Stock Type* is automatically set to **Unrestricted-Use**, and the *Special Stocks* is **None**.

Unrestricted-Use
None

Item 01

Material Storage Location / Stock Type Additional Information

Material:*

TRWA1000

Quantity / Unit:*

500.000

EA

Storage Location / Stock Type

Plant:*

Plant Dallas

Storage Location:*

Semi-Fin. Goods

Stock Type:

Unrestricted-Use

Special Stocks:

None

Supplier:

Click on **Apply and New** to accept your entries and, at the same time, to be able to specify a new item. The system confirms the transfer of the item.

Item 01 is applied.

Now repeat the procedure for the other components of the bike **DXTR3###**.

Material	Quantity	Unit	Plant	SLoc
TRFR3### (Touring-Frame-Red)	500	EA	DL00	RM00

TRFR3###

DGAM1### (Derailleur Gear Assembly)	500	EA	DL00	RM00
TRSK1### (Touring Seat Kit)	500	EA	DL00	RM00
TRHB1### (Touring Handlebar)	500	EA	DL00	RM00
PEDL1### (Pedal Assembly)	500	EA	DL00	RM00
CHAN1### (Chain)	500	EA	DL00	RM00
BRKT1### (Brake Kit)	500	EA	DL00	RM00
WDOC1### (Warranty Document)	500	EA	DL00	RM00
PCKG1### (Packaging)	500	EA	DL00	RM00

DGAM1###
TRSK1###
TRHB1###
PEDL1###
CHAN1###
BRKT1###
WDOC1###
PCKG1###

As soon as you create the last item, confirm it with **Apply** to automatically return to the goods receipt posting. There you will see all created items.

Items

☐	Item	Material	Quantity / Unit	Plant	Storage Location	Stock Type
☐	01	TRWA1000	500.000 EA	Plant Dallas	Semi-Fin. Goods	Unrestricted Use
☐	02	TRFR3000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	03	DGAM1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	04	TRSK1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	05	TRHB1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	06	PEDL1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	07	CHAN1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	08	BRKT1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	09	WDOC1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use
☐	10	PCKG1000	500.000 EA	Plant Dallas	Raw Materials	Unrestricted Use

Note If you have forgotten an item, you can add further items by clicking on **Create**. Furthermore, you can also correct entries if necessary.

Secure your goods receipt with **Post**. The SAP system will assign a unique number to the goods receipt and issue an associated message.

✓ Success

1 document(s) created:
Material Document 4900032016/2022

OK

Material Document
Number

Confirm the success message with **OK**.

Click **SAP** to return to the SAP Fiori Launchpad.

☐

Step 11: Issue Goods to Production Order

Task Issue goods to a production order.

Time 10 min

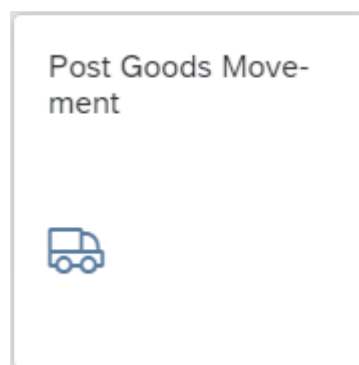
Short Description Now that all necessary components are on stock issue them to your production order in precise quantity.

Name (Position) Sanjay Datar (Warehouse Employee)

The goods issue process is fully defined in the production order, BOM, and routing. The quantities and the materials are reserved for this specific production order, they will be withdrawn with reference to the order number and will be used to assign actual costs to the production order for managerial accounting purposes.

To issue goods to a production order, use the app *Post Goods Movement* in the *Production Planning and Execution* area.

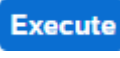
Post Goods Movement



Make sure that **Goods Issue** and **Order** are selected in the drop-down menus.

The *Document Date* as well as the *Posting Date* should be already set to the current date. The *Movement Type* should be set to 261 (GI for order).

Enter your noted **production order** number.

Alternatively, click the value help icon  in the order field. In the Order Number popup (1), use the icon on the far right  to display a list of all tabs. Select the tab *Process Orders using the Info System*. On this tab, enter your material **DXTR3###** in the *Material* field and click **Execute** . Select your order and accept it with .

Process Orders using
the Info System
DXTR3###

Once you have found or entered your production order number, press  to load the order details.

Note Goods issues posting for the required components is another milestone in the production order process. The following functions are performed when a GI for the components of the production order is posted:

- Storage-location-specific update of the stock and consumption fields
- Reduction of the reservation (for planned withdrawal)
- Update of costs for unplanned withdrawals
- Determination of actual costs (valuation) and order update
- Consumption update
- Generation of material and accounting documents (FI and CO documents)
- Creation of material document.
- Creation of accounting document
- Creation of controlling document
- Printing of GI document

The goods issues posting is controlled through a movement type (261), to which each posting refers. This can take place manually or automatically.

An itemized list will appear. It lists all the materials and their respective quantities that need to be issued to your order. You need to tell the system what Storage Location the materials should be withdrawn from. For the Touring Aluminum Wheel Assembly (TRWA1###), enter **SF00** (Semi-finished goods) and for all other materials **RM00** (Raw materials) in the SLoc fields. Before pressing Enter compare your screen with the one shown below. Notice that your quantity could be different.

SF00

RM00

Flag each item with **OK**. If you can't flag the first item please close the detail view with a click on  under the line item list.

OK

Line	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	S...	SLoc
1	Touring Frame-Red	☐	☒	124	EA		RM00
2	Derailleur Gear Assembly	☐	☒	124	EA		RM00
3	Touring Seat Kit	☐	☒	124	EA		RM00
4	Pedal Assembly	☐	☒	124	EA		RM00
5	Touring Handle Bar	☐	☒	124	EA		RM00
6	Chain	☐	☒	124	EA		RM00
7	Brake Kit	☐	☒	124	EA		RM00
8	Warranty Document	☐	☒	124	EA		RM00
9	Packaging	☐	☒	124	EA		RM00
10	Touring Aluminum Wheel Assembly	☐	☒	248	EA		SF00

Click on  and record the material document number.

Material Document
Number

 Material document 4900032017 posted [View Details](#)

Click  to return to the SAP Fiori Launchpad.



Step 12: Review Production Order Status

Task Review the production order status.

Time 10 min

Short Description Review the current production order with respect to the status of the order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

To display the production order, use the app *Manage Production Orders* in the *Production Planning and Execution* area.

Manage Production Orders - Orders



When you open the app for the first time, you will see a welcome message telling you to select an area of responsibility. Confirm this with OK, another popup will appear.

Area of Responsibility	
Production Supervisor	>
Work Center	>
OK	

Choose *Production Supervisor*. No Plant is currently assigned. Click on + an choose the **Plant Dallas**. Confirm your selection with Select.

Area of Responsibility (Supervisor)

Add Plant and Production Supervisor 

Plant Dallas (DL00) 

DL Production Scheduler (000)

OK Cancel

Confirm your selection with  now click on *Work Center*.

Click on  again and choose the *Work Center* **DL Assembly (ASSY1000)**, **DL Inspection (INSP1000)** and **DL Packaging (PACK1000)** from your

Plant Dallas and click on .

Area of Responsibility (Work Center)

Add Plant and Work Center 

Plant Dallas (DL00) 

DL Assembly (ASSY1000)

Plant Dallas (DL00) 

DL Inspection (INSP1000)

Plant Dallas (DL00) 

DL Packaging (PACK1000)

OK Cancel

Confirm your selection with , and press  to leave the selection or the area of Responsibility.

You will receive an overview of all existing orders. Depending on the progress of your course, there may be several production orders with different processing statuses.

SAP Manage Production Orders									
Standard									
Search Station: Delivered < More Issue Type: No Filter < Delay Duration: < 0 Hours Order: Material: Scheduled Start: Adapt Filters <									
Orders (4)									
☐	Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues	
☐	1000000	DXTR0000 Deluxe GPS-Bike Computer Silver White	0 EA	Delivered	Tue, Sep 21, 2022 04:36	Wed, Sep 22, 2022 03:30	Move to transport (0080)		
☐	1000000	DXTR0000 Deluxe Touring Bike (red)	0 EA	Delivered	Wed, Sep 15, 2022 04:57	Wed, Sep 15, 2022 04:57	Move to storage (0100)		
☐	1000000	DXTR0000 Deluxe GPS-Bike Computer Royal Blue	0 EA	Delivered	Fri, Sep 9, 2022 09:31	Fri, Sep 9, 2022 09:30	Move to transport (0080)		
☐	1000000	DXTR0000 Deluxe Touring Bike (red)	124 EA	Released	Sun, Oct 24, 2022 04:09	Sat, Oct 30, 2022 17:00	Material staging (0000)		

In the field *Material* enter your material **DXTR3###** and choose  to display your order.

DXTR3###

SAP Manage Production Orders									
Standard									
Search Station: Delivered < More Issue Type: No Filter < Delay Duration: < 0 Hours Order: Material: Deluxe Touring Bike (red) DXTR3 EA Scheduled Start: Adapt Filters <									
Orders (1)									
☐	Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues	
☐	1000000	DXTR0000 Deluxe Touring Bike (red)	124 EA	Released	Sun, Oct 24, 2022 04:09	Sat, Oct 30, 2022 17:00	Material staging (0000)		

The tabular overview already provides you with various information about your order, such as the current status and the current processing status.

For further information select the entry. You will be forwarded to the details of the production order.

SAP Manage Production Orders			
1000000		Edit Order Display Configuration	
Material:	Status:	Person Responsible:	Quantity:
DXTR3000 (Deluxe Touring Bike (red))	Released	DL MRP Controller	124 EA
Issues Order Information Components Order Schedule Confirmation Inspection			
Order Information			
General		Dates/Times	Quantities
Production Plant: DL00		Scheduled Start: Sat, Jul 23, 2022, 10:12	Total Quantity: 124
MRP Area: DL00 Plant Dallas		Scheduled End: Sat, Jul 30, 2022, 17:00	Confirmed Yield: 0
Planning Plant: DL00 Plant Dallas		Planned Start : Fri, Jul 22, 2022, 00:00	Confirmed Scrap: 0
Production Version: 0001 Generated Version 0001		Planned End: Mon, Aug 1, 2022, 00:00	GR Quantity: 0
MRP Controller: 000		Scheduled Release: Fri, Jul 22, 2022	Open Quantity: 124
Processing: Sequential			

Click on the status **Released** for more information. You can see that your production order is precosted and a settlement rule has been created.

1000000

Material:

DXTR3000 (Deluxe Touring Bike (red))

Status:

Released

Status Overview

Released

Precosted

Settlement Rule Created

Issues

Order Information

Components

Order Schedule

Confirmation

Inspection

Now click on the *Components* tab. The screen scrolls to the corresponding position.

Components

AllMissing

Material	Quantity	Coverage	Backflush	Component Scrap	Requirement Date/Time	Storage Location
TRFR3000 Touring Frame-Red	Total Quantity: 124 E Open Quantity: 0 EA	124 EA	No	0.00 %	Sat, Jul 23, 2022 11:45	
TRSK1000 Touring Seat Kit	Total Quantity: 124 E Open Quantity: 0 EA	124 EA	No	0.00 %	Sat, Jul 23, 2022 11:45	
TRHB1000 Touring Handle Bar	Total Quantity: 124 E Open Quantity: 0 EA	124 EA	No	0.00 %	Sat, Jul 23, 2022 14:05	
DGAM1000 Derailleur Gear Assembly	Total Quantity: 124 E Open Quantity: 0 EA	124 EA	No	0.00 %	Sun, Jul 24, 2022 09:44	

In the last task, you posted the goods issue for the production order. In the production order, you now see that there are no more open quantities for this order.

Click  to return to the SAP Fiori Launchpad.



Step 13: Confirm Production Completion

Task Confirm production order completion.

Time 10 min

Short Description Confirm completion for your production order.

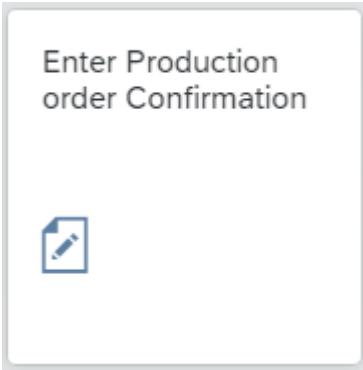
Name (Position) Michael Brauer (Shop Floor Worker 4)

When the assembly has been completed for the current production order, we need to confirm that certain procedures and activities have been completed and record the quantity of the finished product that has been manufactured.

Production completion

To confirm production completion, use the app *Enter Production order Confirmation* in the *Production Planning and Execution* area.

Enter Production order Confirmation



Enter your **production order** number and click on **Continue** or Enter.

Production order number

Alternatively, use the Input-Help-Symbol  and switch in the popup with  to the tab *production orders by material and routing*, enter your material number **DXTR3##** there and search for your order.

Check if **Final Confirmation** and **Clear Open Reserv.** is already checked in the *Confirmation Type* section.

Final Confirm.
Clear Reservation
Amount

 Confirmation of Production Order Enter : Actual Data

 More 

Order:

Status: REL PRC GMPS MACM SETC

Material:

Material Descr.:

Confirmation Type

Partial confirmation: ☐

Clear Open Reservs.: ☒

Final Confirmation: ☒

Autom. Final Conf.: ☐

Furthermore, in the *Actual Data* tab, the quantity of bicycles that you should produce for this order should already be entered in the *Yield Quantity* field. Change the *Start Execution* to **1 hour earlier** than the pre-set time.

1 hour earlier

Actual Data

	Curr. t/b Conf.	Unit	Confirmed to Date	Planned t/b Conf. Unit
Yield Quantity:	124	EA	0	124 EA
Scrap Quantity:			0	0
Rework Quantity:			0	
Reason for Var.:				
Personnel no.:				

	To Be Confirmed	Confirmed to Date	Planned t/b Conf.
Start Execution:	06/26/2022 08:30:04		07/23/2022
Finish Execut.:	06/26/2022 09:30:04		07/30/2022
Posting Date:	06/26/2022		

Confirm. Text:

Long Text exists: ☐

Save your entries with . You will get a confirmation from the system.

 Confirmation of order 1000000 saved

Note When the confirmation is saved, labor costs for the order are calculated automatically. The quantity yield also establishes the parameters for the goods receipt into Inventory.

Click  to return to the SAP Fiori Launchpad.



Step 14: Review Production Order Status

Task Review the production order status.

Zeit 10 Min.

Short Description Review the current production order with respect to the status of the order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

To display the production order, use the app *Manage Production Orders* in the *Production Planning and Execution* area.

Manage Production Orders



In the field *Material* enter your material **DXTR3###** and choose **Go** to display your order.

DXTR3###

SAP Manage Production Orders

Standard

Search

Status: 7 Items

Issue Type: No Filter

Delay Duration: >= 0 Hours

Order:

Material:

Scheduled Start:

Adapt Filters (3) Go

Orders (1)

Change Dates and Quantities Edit Release Read Master Data

Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues
☐	1000000 DXTR3000 Deluxe Touring Bike (red)	124 EA	Confirmed	Sun, Jun 26, 2022 08:30	Sun, Jun 26, 2022 09:30	Move to storage (0110) 124 of 124	

As you can see, the status of your production order has changed from *Released* to *Confirmed*. Furthermore, the processing status is now at *Move to storage*.

For more information, select the entry, you will be redirected to the details of the production order. Click on the *Confirmation* tab to go to the related area.

Confirmation

Order Confirmations							
Confirmation	Confirmation Count	Reversed	Reversal	Reversed Count	Operation	Confirmed By	Confirmed
116	1	No	No			LEARN-400	Sun, Jun 26 09:30 >
Final Confirmation: No							
Quantity: 124 EA							
Yield: 124 EA							
Scrap: 0 EA							
Rework: 0 EA							

An order confirmation is now available. You can see that the complete quantity of your production order has been confirmed and that there is no scrap.

After the confirmation, the goods receipt now needs to take place to complete the order.

Click  to return to the SAP Fiori Launchpad.



Step 15: Receive Goods from Production Order

Task Post a goods receipt from production order.

Time 15 min

Short Description Post a goods receipt from your production order.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Receive the completed products into finished goods inventory. Check the quantity proposed against the quantity specified in the production order and the quantity specified during confirmation. If there are any discrepancies, the system will decide if an error or warning message should be generated, depending upon the deviation identified.

Goods receipt

To post the goods receipt, use the app *Post goods receipt for production order*.

Post goods receipt for production order



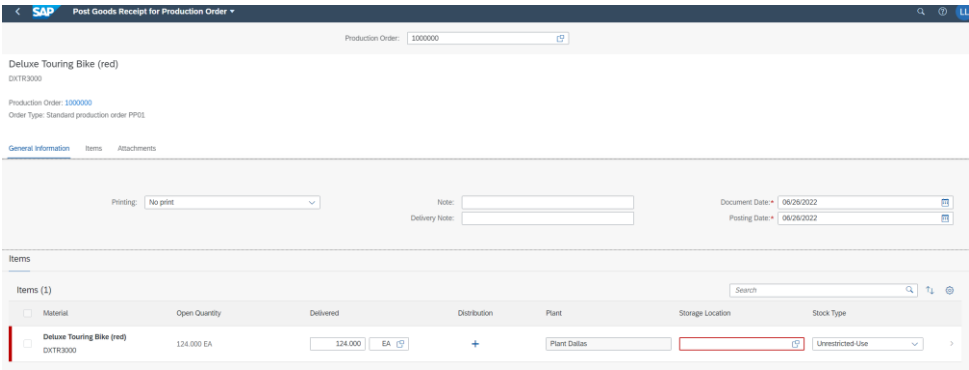
Enter your noted **production order** and press Enter.

Production order

As an alternative use the Input-Help-Search  and enter your material **DXTR3###** and click . Then select your production order from the result list.

DXTR3###

Your production order is loaded and displayed.



Add the storage location **FG00** for end products, all other settings can be accepted.

Items						
Items (1)						
Material	Open Quantity	Delivered	Distribution	Plant	Storage Location	Stock Type
☒ Deluxe Touring Bike (red) DKTR3110	101.000 EA	101.000 EA	+	Plant Dallas	Finished Goods	Unrestricted-Use

Save your goods receipt with . The SAP system will assign a unique number to the goods receipt and issue a corresponding message.



Success

1 document(s) created:

 Material Document 5000000013/2022

OK

Material Document
Number

Furthermore, this updates the current value of the material produced to the production order.

Confirm the successmessage with .

Click  to return to the SAP Fiori Launchpad.

Step 16: Review Costs Assigned to Production Order

Task View the corresponding costs assigned to your production order.

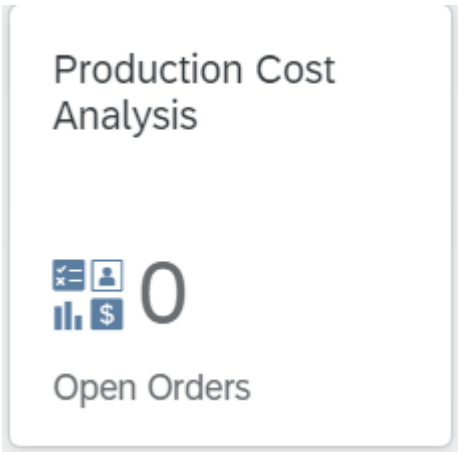
Zeit 5 Min.

Description Review all costs associated with your production order.

Name (Position) Jamie Shamblin (Controller)

To view the cost of a manufacturing job, use the *Production Cost Analysis* app.

Production Cost
Analysis



In the *Material* field enter **DXTR3###** and change the *Order Status* from **Open** to **Closed**

DXTR3###
Closed

Profit Center:

Plant:

Order Type:

Material:

Order:

DXTR3110

Plan Category: *

Order Status: *

Period From: *

Period To: *

Clear

Adapt Filters (5)

Go

Production Order Standar...

Closed

009/2021

009/2021

Click on **Go** to start the search. Your recently completed production order will be displayed.

Order Analysis

Cost Component Group

Exception Status	Order	Material	Tgr DR/Act DR Vari: =	Target Cost Debit	Act Cost Debit	Total Act Cost	Order Status
(1)	1000000	DXTR3000 (Deluxe Touring Bike [...	-17,484.10 USD	108,067.65 USD	90,583.55 USD	-190.65 USD	Closed

This overview lists the summed target and actual costs and shows any variances. Click on **>** at the end of the entry to open the cost details.

SAP

Order Cost Detail

Search

Help

LL

PP01: 1000000

Exception Status

Order Balance

No Defined Exception Rule

-190.65 USD

GENERAL INFORMATION

COST DETAILS

Material:

DXTR3000 (Deluxe Touring Bike (red))

Planned Quantity:

124 EA

From Period:

006/2022

Plan Category

Production Order Standard Cost

Total Actual Quantity:

124 EA

To Period:

011/2022

COST DETAILS

Standard

Target/Actual by G/L Account

G/L Account	Total Target Cost	Total Actl Cost	Tgt/Act Cost/Var	Actual Quantity	Origin	
Group Name:						
7520000 (Settle. ...)	-90,774.20 USD	-90,774.20 USD	0.00 USD	-124 EA	DL00/DXTR3000	
Group Name: Human Resources						
8000000 (Labor)	3,089.25 USD	3,101.55 USD	12.30 USD	62.031 HR	NAPR1000/LABOR	Le
Group Name: Material Overhead						
8015000 (Material...)	17,496.40 USD	0.00 USD	-17,496.40 USD	0	NAPR1000	
Group Name: Raw Materials						
5001000 (RM Con...)	8,680.00 USD	8,680.00 USD	0.00 USD	124 EA	DL00/BRKT1000	
5001000 (RM Con...)	1,240.00 USD	1,240.00 USD	0.00 USD	124 EA	DL00/CHAN1000	
5001000 (RM Con...)	9,300.00 USD	9,300.00 USD	0.00 USD	124 EA	DL00/DGAM1000	
5001000 (RM Con...)	434.00 USD	434.00 USD	0.00 USD	124 EA	DL00/PCKG1000	
5001000 (RM Con...)	5,580.00 USD	5,580.00 USD	0.00 USD	124 EA	DL00/PEDL1000	
5001000 (RM Con...)	24,800.00 USD	24,800.00 USD	0.00 USD	124 EA	DL00/TRFR3000	
5001000 (RM Con...)	3,100.00 USD	3,100.00 USD	0.00 USD	124 EA	DL00/TRHB1000	
	17,293.45 USD	-190.65 USD	-17,484.10 USD	1,302.031 *		

Now that the finished products have been received in the Inventory, the Manufacturing Output Settlement Variance has been added. How is this figure calculated by the system?

Click  to return to the SAP Fiori Launchpad.



Step 17: Settle Costs of Production Order

Task Settle costs of your production order.

Time 20 min

Short Description Settle the costs of your production order. The costs are temporarily captured in the production order and they need to be assigned to an appropriate cost object. Compare the actual costs to the planned costs to identify any deviations or potential problems in this regard.

Name (Position) Jamie Shamblin (Cost Accountant)

To settle costs of a production order, use the app *Run Actual Settlement - Order - Single* in the *Production Planning and Execution* area.

Run Actual Settlement - Order - Single



If you have to input the Controlling Area, enter **NA00**, and click on **Continue**.

NA00

Enter your **production order number**, alternatively search for it with your material DXTR3### in the Input-Help. In the *Parameters* section enter *Settlement- and posting period* as **current month** (e.g., 006 for June). Enter as *Fiscal Year* the **current year**. Make Sure that **Test Run** is checked in the *Processing Options* section.

Order number
current month
current year
Test Run

Controlling Area: NA00

* Order:

Parameters

* Settlement Period:

* Fiscal Year:

* Processing Type: 1 Automatic

Posting period:

Asset Value Date:

Processing Options

☒ Test Run

☐ Check Trans. Data

Execute

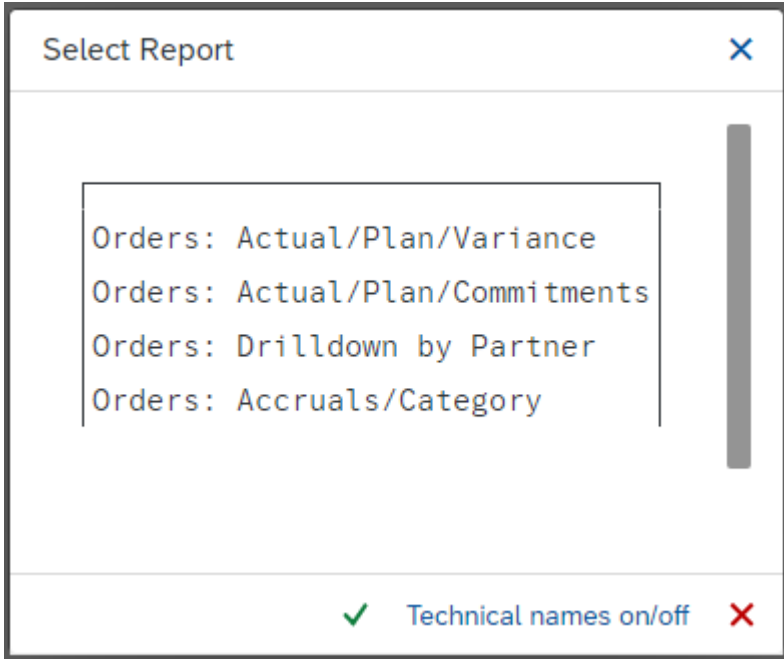
< SAP Actual Settlement: Order Basic list		LL ? Q	
≡ 📄 More ▾		Find 🔍 🏠 Exit	
Selection			
Selection Parameters		Value	Name
Order		1000000	Deluxe Touring Bike (red)
Period		006	
Posting Period		006	
Fiscal Year		2022	
Processing Type		1	Automatic
Posting Date		06/30/2022	
Controlling Area		NA00	Global Bike North America
Currency		USD	United States Dollar
Value Date		06/30/2022	
.....			
Processing Options			
Selection Parameters		Value	
Execution Type		Settlement Executed	
Processing Mode		Test run	
.....			
Processing completed with no errors			
.....			
Statistics			
Processing Category		Number	
Settlement Executed		1	
No Change			
Not Relevant			
Inappropriate Status			
Error			

Detail list - Settled values				
Senders	Short text: Sender	Receiver	Val/COArea CrCy	Additional information
ORD 1000000	Deluxe Touring Bike (red)	MAT DL00/DXTR3000	190.65-	
			• 190.65-	

Menu Bar

More ► Environment ► Report

The *Select Report* pop-up will appear.



Click on **Actual/Plan/Variance** and confirm your selection with . A corresponding report group is generated and displayed.

Actual/Plan/Variance

Orders: Actual/Plan/Variance		Date: 06/26/2022 16:41:43	Page: 2 / 2
Order/Group	1000000	000001000000	
Fiscal year	2022		
Period	1 - 6		

Cost Elements	Actual	Plan	Var. (Abs.)	Var. (%)
5001000 RM Consumpt Expense	59,458.00		59,458.00	
5004000 SF Consumpt Expense	28,024.00		28,024.00	
8000000 Labor	3,101.55		3,101.55	
8015000 Material Overhead				
* Costs	90,583.55		90,583.55	
7520000 Settle. Manuf Output	90,774.20-		90,774.20-	
* Deliveries to Stock	90,774.20-		90,774.20-	
** Balance	190.65-		190.65-	

The test run is now complete and the *actual settlement* should now be performed.

Click on  to go back. Confirm the Message to leave the report with . Afterwards press  two times to get back to the entry screen.

Testlauf

Deselect Test Run and again click on **Execute**. In contrast to the previous run, you can now see in the *Processing Options* area that this was an update run that was *completed with no errors*.

Processing Options	
Selection Parameters	Value
Execution Type	Settlement Executed
Processing Mode	Update run
Processing completed with no errors	

Open the report **Actual/Plan/Variance** again by clicking on  first and then use the menu bar path **More ▶ Environment ▶ Report** and select the option **Actual/Plan/Variance**.

Actual/Plan/Variance

Orders: Actual/Plan/Variance		Date: 06/26/2022 16:42:52	Page: 2 / 2
Order/Group	1000000	000001000000	
Fiscal year	2022		
Period	1 - 6		

Cost Elements	Actual	Plan	Var. (Abs.)	Var. (%)
5001000 RM Consumpt Expense	59,458.00		59,458.00	
5004000 SF Consumpt Expense	28,024.00		28,024.00	
8000000 Labor	3,101.55		3,101.55	
8015000 Material Overhead				
* Costs	90,583.55		90,583.55	
7520000 Settle. Manuf Output	190.65		190.65	
* Settled Costs	190.65		190.65	
7520000 Settle. Manuf Output	90,774.20-		90,774.20-	
* Deliveries to Stock	90,774.20-		90,774.20-	
** Balance				

You can see that the costs have now been settled.

Click  to return to the SAP Fiori Launchpad.



PP Challenge

Learning Objective Understand and perform an integrated manufacturing process.

Time 60 min

Motivation After you have successfully worked through the *Production Planning and Execution* case study you should be able to solve the following challenge on your own.

Scenario In this challenge you should create sales and operations plan (SOP) for the product group (product family) Mountain bikes. Take into consideration that the materials of the product group have to be assigned to the strategy group. Therefore, enter manually the following sales figures:

Period	Sales (volume)
Current month + 2	150
Current month + 3	175
Current month + 4	200
Current month + 5	85
Current month + 6	90
Current month + 7	115

In addition, you must post the correct goods for Material ORMN1### in the storage location in order to be able to produce and settle costs afterwards.

Task Information Since this task is based on the *Production Planning and Execution* case study you can use it as guidance. However, it is recommended that you solve it without any help in order to test your acquired knowledge.

